Challenge Problems Week 1

Ryan's Favorite Experiment

UCQ Education Team

Light was the first thing to be described as "quantum". Let's use quantum mechanics to uncover a counterintuitive result.

A photon (quantized light wave) can be polarized along two axes, *vertical* and *horizontal*.

$$|\psi\rangle = a\,|H\rangle + b\,|V\rangle \qquad\qquad |a|^2 + |b|^2 = 1$$

Problem 1. Verify that the pairs of states

$$|D\rangle = \frac{1}{\sqrt{2}}\,|H\rangle + \frac{1}{\sqrt{2}}\,|V\rangle \qquad |A\rangle = \frac{1}{\sqrt{2}}\,|H\rangle - \frac{1}{\sqrt{2}}\,|V\rangle$$

("diagonal" and "anti-diagonal" polarization) and

$$|R\rangle = \frac{1}{\sqrt{2}}\,|H\rangle + \frac{i}{\sqrt{2}}\,|V\rangle \qquad |L\rangle = \frac{1}{\sqrt{2}}\,|H\rangle - \frac{i}{\sqrt{2}}\,|V\rangle$$

("right" and "left-handed circular" polarization) both form different orthonormal bases for the Hilbert space.

A *polarizer* performs a measurement operation in the direction of the polarizer. It will let a photon through if the polarization of the photon is measured to be in the same direction as the polarizer and will block the photon if the polarization is measured to be in a perpendicular direction. Consider photons polarized in any orientation

$$|\psi\rangle = \cos(\theta)\,|H\rangle + \sin(\theta)\,|V\rangle$$

Problem 2. Show that $\langle\psi|\psi\rangle = 1$, so $|\psi\rangle$ is a valid quantum state. What's the probability of measuring $|H\rangle$? What about $|V\rangle$?

Problem 3. Consider now a polarizer oriented at the angle θ above the horizontal. What's the probability that an $|H\rangle$ photon passes through? (Hint: tilt your head!)

Consider now an unpolarized light source emitting photons into two polarizers. The photons first pass through a $|H\rangle$-oriented polarizer and then a $|V\rangle$-oriented polarizer.

Problem 4. Show that no matter the initial state

$$|\psi\rangle = a\,|H\rangle + b\,|V\rangle$$

the photons have a 0% chance of passing thorough both polarizers.

Problem 5. Now add a third polarizer (which **only removes photons**) in between the two polarizers oriented at an angle θ. What proportion of photons that pass through the first polarizer will pass through the last polarizer (as a function of θ)? Is it nonzero?